

1. Write a program to display:

- value

- Type

- memory address

for a variable using appropriate built in functions.

Ans:

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2. What is the difference between

$a = 10$

$b = 10$

and

$a = [10]$

$b = [10]$

Explain using id.

Ans:

In first case:

$a = 10$

$b = 10$

In this case both a and b will refer the same reference id in

heap memory. This is immutable and it will save same id.

<i>Heap Memory</i>	
<i>a</i>	<i>104009</i>
<i>b</i>	<i>104009</i>

In case of:

a = [10]

b = [10]

a and b will refer the different id in heap memory. This is mutable and use different id.

<i>Heap Memory</i>	
<i>a</i>	<i>0001</i>
<i>b</i>	<i>0002</i>

3. What does the id() function return? Is the value returned by id() same for two different variable holding the same value?

Ans:

The id() function returns a unique value/number in the memory.

If the value is same it will return same reference id in the memory or it will create different id if values are different.

4. What is the purpose of `getsizeof()`? Why is the memory size different for different data types?

Ans:

The `getsizeof()` function shows the total memory space occupied (in bytes) by an object.

Difference between datatypes decides memory size of different data types.

5. Predict the output:

a = 10

b = 10

print(id(a) == id(b))

Explain why this happens.

Ans:

Output will be as follows:

True (e.g. 140715993564568 == 140715993564568)

Since integer uses same heap memory because value of variable is same. This will show same reference id for variable 1 and variable 2 (a and b).

6. Write a program that accepts two numbers from user and prints their:

- Addition
- Subtraction
- Multiplication
- Division

Ans:

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7. Why does the `input()` function always return a string?

How can you convert it to another data type?

Ans:

Because the `input()` function designed to read text entered by the user.

This can be converted like below:

`int(input("10"))` → 10

`float(input("10.5"))` → 10.5

8. Predict the output:

`x = input("Enter number ")`

```
print(type(x))
```

Explain the reason.

Ans:

Output will be:

<class 'str'>

Because input() function designed to read text in string format entered by the user.

9. Write program to take user's name and age and display:

Hello <name>, you will turn <age+1> next year.

Ans:

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10. What will be the output:

`print("10" + "20")`

`print(10 + 20)`

Explain the difference.

Ans:

Output will be:

1020

30

In first output it is string concatenation.

In second output it is integer addition.